

Sources (proxies)

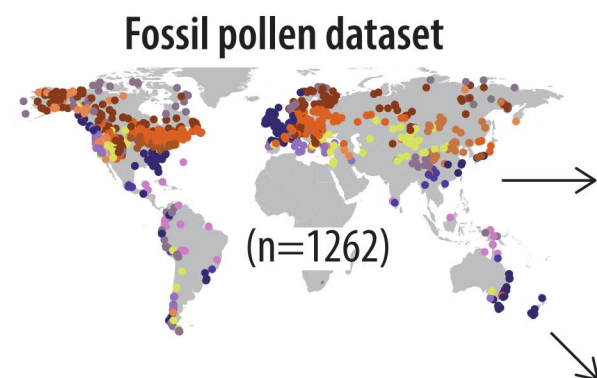
Variables

Inference

Spatial units

HVARPART analysis

Predictor importance (ratio)



Pollen assemblage
properties

Human impact
events



SPDs



Database of
Archeological
¹⁴C

CHELSA-
TraCE21k
palaeoclimate

MAT °C
Cold °C
Summer prec
Winter prec

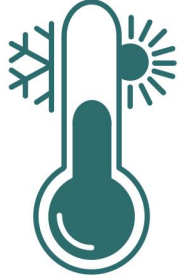
Past vegetation
properties



Human impact



Climate



Samples
in records

Samples across
continental groups

Samples across
regions and
continental groups

General trends across
regions and
continental groups

Model changes in properties

Response
Matrix of pollen-
assemblage
properties

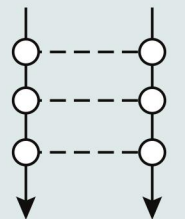
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Predictor
Human
impact

+

Predictor
Model changes
in properties

Redundancy analysis (RDA) with scaling of variables



Model regional changes in interrelationships

PCA per time
slice/climate zone
Matrix of pollen-
assemblage
properties

→

Pairwise
Procrustes
sum of squares
distance
matrix

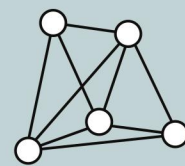
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General
predictor
trends
Human
impact

+

General
predictor
trends
Matrix of
climate variables

Distance-based RDA with sum-of-squares (m²)



Total predictor
importance

$$\text{Human impact} = \frac{\text{Human impact}}{(\text{Human impact} + \text{Climate})}$$

$$\text{Climate} = \frac{\text{Climate}}{(\text{Climate} + \text{Human impact})}$$